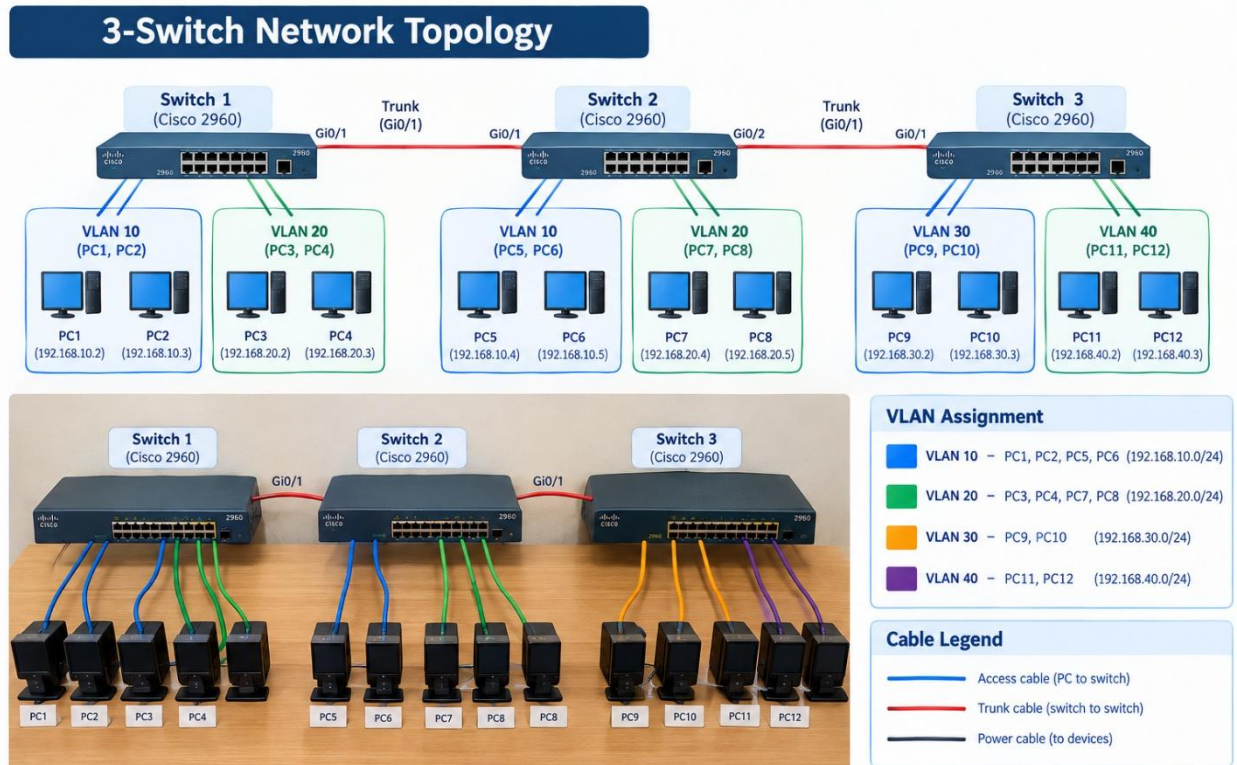


VPN AND PHYSICAL NETWORK LABS

1. Set up a 3-switch physical network topology using actual switches and Ethernet cables, and draw a labeled diagram of your setup showing all connections.

->



2. Configure VLANs on each switch so that at least two VLANs exist per switch, and assign at least two ports to each VLAN; take screenshots of your configuration commands or switch console output.

->**Switch 1 — VLAN 10 and VLAN 20**

Switch1> enable

Switch1# configure terminal

Switch1(config)# vlan 10

Switch1(config-vlan)# name SALES

Switch1(config-vlan)# exit

Switch1(config)# vlan 20

```
Switch1(config-vlan)# name HR
```

```
Switch1(config-vlan)# exit
```

```
Switch1(config)# interface range fa0/1-2
```

```
Switch1(config-if-range)# switchport mode access
```

```
Switch1(config-if-range)# switchport access vlan 10
```

```
Switch1(config-if-range)# exit
```

```
Switch1(config)# interface range fa0/3-4
```

```
Switch1(config-if-range)# switchport mode access
```

```
Switch1(config-if-range)# switchport access vlan 20
```

```
Switch1(config-if-range)# exit
```

```
Switch1(config)# end
```

```
Switch1# show vlan brief
```

Expected output:

VLAN Name	Status	Ports
-----------	--------	-------

10 SALES	active	Fa0/1, Fa0/2
----------	--------	--------------

20 HR	active	Fa0/3, Fa0/4
-------	--------	--------------

Switch 2 — VLAN 10 and VLAN 20

```
Switch2> enable
```

```
Switch2# configure terminal
```

```
Switch2(config)# vlan 10
```

```
Switch2(config-vlan)# name SALES
```

```
Switch2(config-vlan)# exit
```

```
Switch2(config)# vlan 20
```

```
Switch2(config-vlan)# name HR
```

```
Switch2(config-vlan)# exit
```

```
Switch2(config)# interface range fa0/1-2
```

```
Switch2(config-if-range)# switchport mode access
```

```
Switch2(config-if-range)# switchport access vlan 10
```

```
Switch2(config-if-range)# exit
```

```
Switch2(config)# interface range fa0/3-4
```

```
Switch2(config-if-range)# switchport mode access
```

```
Switch2(config-if-range)# switchport access vlan 20
```

```
Switch2(config-if-range)# exit
```

```
Switch2(config)# end
```

```
Switch2# show vlan brief
```

Expected output:

VLAN Name	Status	Ports

10 SALES	active	Fa0/1, Fa0/2
20 HR	active	Fa0/3, Fa0/4

Switch 3 — VLAN 30 and VLAN 40

```
Switch3> enable
```

```
Switch3# configure terminal
```

```
Switch3(config)# vlan 30
```

```
Switch3(config-vlan)# name IT
```

```
Switch3(config-vlan)# exit
```

```
Switch3(config)# vlan 40
```

```
Switch3(config-vlan)# name MANAGEMENT
```

```
Switch3(config-vlan)# exit
```

```
Switch3(config)# interface range fa0/1-2
```

```
Switch3(config-if-range)# switchport mode access
```

```
Switch3(config-if-range)# switchport access vlan 30
```

```
Switch3(config-if-range)# exit
```

```
Switch3(config)# interface range fa0/3-4
```

```
Switch3(config-if-range)# switchport mode access
```

```
Switch3(config-if-range)# switchport access vlan 40
```

```
Switch3(config-if-range)# exit
```

```
Switch3(config)# end
```

```
Switch3# show vlan brief
```

Expected output:

VLAN Name	Status	Ports
-----------	--------	-------

30	IT	active	Fa0/1, Fa0/2
----	----	--------	--------------

40	MANAGEMENT	active	Fa0/3, Fa0/4
----	------------	--------	--------------

VLAN assignment summary

Switch	VLAN	Name	Ports
Switch 1	10	SALES	Fa0/1–Fa0/2

Switch	VLAN	Name	Ports
Switch 1	20	HR	Fa0/3–Fa0/4
Switch 2	10	SALES	Fa0/1–Fa0/2
Switch 2	20	HR	Fa0/3–Fa0/4
Switch 3	30	IT	Fa0/1–Fa0/2
Switch 3	40	MANAGEMENT	Fa0/3–Fa0/4

3. Enable trunking on the links connecting the switches, and verify trunk status using appropriate show commands; include screenshots of your verification.

-> **Trunking Configuration — 3-Switch Network**

For the physical setup from the previous step, configure the **switch-to-switch links as 802.1Q trunks**. Assume:

Switch 1 Gi0/1 ————— Gi0/1 Switch 2

Switch 2 Gi0/2 ————— Gi0/1 Switch 3

1. Switch 1 → Switch 2

On **Switch 1**:

```
Switch1> enable
```

```
Switch1# configure terminal
```

```
Switch1(config)# interface gigabitEthernet 0/1
```

```
Switch1(config-if)# switchport mode trunk
```

```
Switch1(config-if)# switchport trunk allowed vlan 10,20,30,40
```

```
Switch1(config-if)# no shutdown
```

```
Switch1(config-if)# exit
```

```
Switch1(config)# end
```

```
Switch1# show interfaces trunk
```

2. Switch 2 → Switch 1 and Switch 3

On **Switch 2**:

Switch2> enable

Switch2# configure terminal

Switch2(config)# interface gigabitEthernet 0/1

Switch2(config-if)# switchport mode trunk

Switch2(config-if)# switchport trunk allowed vlan 10,20,30,40

Switch2(config-if)# no shutdown

Switch2(config-if)# exit

Switch2(config)# interface gigabitEthernet 0/2

Switch2(config-if)# switchport mode trunk

Switch2(config-if)# switchport trunk allowed vlan 10,20,30,40

Switch2(config-if)# no shutdown

Switch2(config-if)# exit

Switch2(config)# end

Switch2# show interfaces trunk

3. Switch 3 → Switch 2

On **Switch 3**:

Switch3> enable

Switch3# configure terminal

Switch3(config)# interface gigabitEthernet 0/1

Switch3(config-if)# switchport mode trunk

Switch3(config-if)# switchport trunk allowed vlan 10,20,30,40

Switch3(config-if)# no shutdown

```
Switch3(config-if)# exit
```

```
Switch3(config)# end
```

```
Switch3# show interfaces trunk
```

Expected verification

Run:

```
show interfaces trunk
```

You should see output similar to:

Port	Mode	Encapsulation	Status	Native vlan
Gi0/1	on	802.1q	trunking	1

And:

```
Port    Vlans allowed on trunk
```

```
Gi0/1   10,20,30,40
```

On **Switch 2**, both trunk ports should appear:

Port	Mode	Encapsulation	Status
Gi0/1	on	802.1q	trunking
Gi0/2	on	802.1q	trunking

Useful additional verification

Run:

```
show interfaces gi0/1 switchport
```

Look for:

Administrative Mode: trunk

Operational Mode: trunk

Operational Trunking Encapsulation: dot1q

Screenshots to include in your practical record

 **Screenshot 1:** Switch 1 showing:

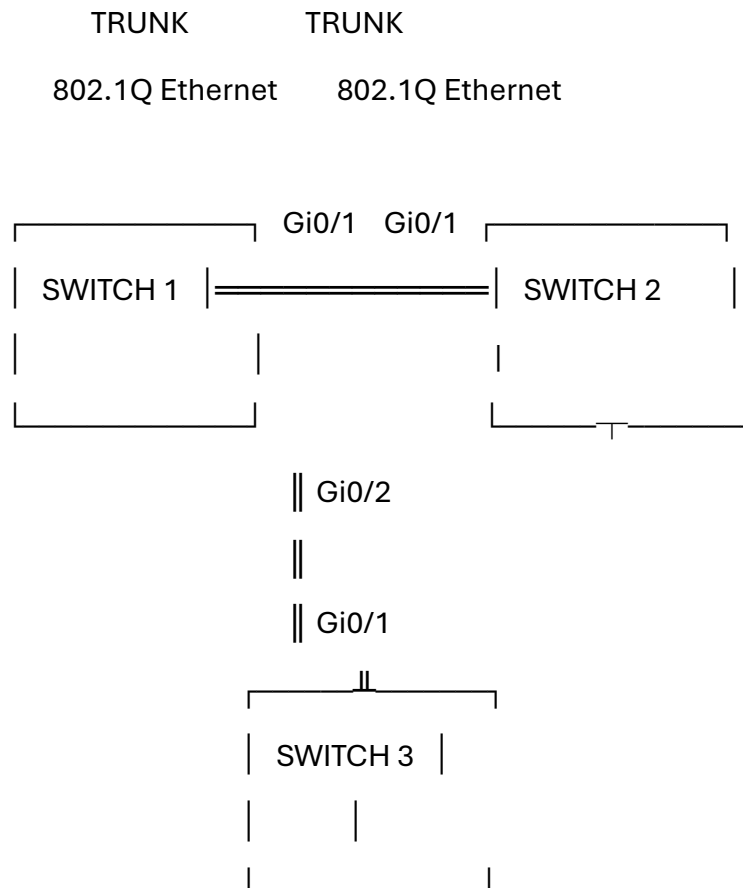
```
show interfaces trunk
```

 **Screenshot 2:** Switch 2 showing both Gi0/1 and Gi0/2 as **trunking**.

 **Screenshot 3:** Switch 3 showing Gi0/1 as **trunking**.

 **Screenshot 4:** Optional show interfaces gi0/1 switchport showing **Administrative Mode: trunk** and **Operational Mode: trunk**.

Physical connection



===== = Trunk Ethernet link

4. Create static routes between the VLANs on your network so devices in different VLANs can communicate, and document the routing commands you used.

->Static Routing Between VLANs

For the 3-switch network, use **router-on-a-stick** for the VLAN gateways and **static routes** between the routers. This allows devices in different VLANs to communicate.

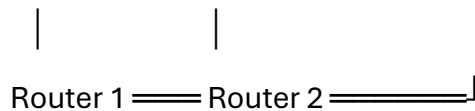
Network addressing

VLAN	Network	Gateway
VLAN 10 – SALES	192.168.10.0/24	192.168.10.1

VLAN	Network	Gateway
VLAN 20 – HR	192.168.20.0/24	192.168.20.1
VLAN 30 – IT	192.168.30.0/24	192.168.30.1
VLAN 40 – MANAGEMENT	192.168.40.0/24	192.168.40.1

Assume the topology is:

PCs → Switch 1 — Switch 2 — Switch 3 → PCs



- **Router 1:** VLAN 10 and VLAN 20
- **Router 2:** VLAN 30 and VLAN 40
- Router 1 ↔ Router 2 uses a separate transit network: 10.0.0.0/30

1. Configure VLAN gateways on Router 1

Connect Router 1's G0/0 to a trunk port on Switch 1.

```
Router1> enable
```

```
Router1# configure terminal
```

```
Router1(config)# interface gigabitEthernet 0/0
```

```
Router1(config-if)# no shutdown
```

```
Router1(config-if)# exit
```

```
Router1(config)# interface gigabitEthernet 0/0.10
```

```
Router1(config-subif)# encapsulation dot1Q 10
```

```
Router1(config-subif)# ip address 192.168.10.1 255.255.255.0
```

```
Router1(config-subif)# exit
```

```
Router1(config)# interface gigabitEthernet 0/0.20
```

```
Router1(config-subif)# encapsulation dot1Q 20
```

```
Router1(config-subif)# ip address 192.168.20.1 255.255.255.0
```

```
Router1(config-subif)# exit
```

Router 1 now acts as the default gateway for VLAN 10 and VLAN 20.

2. Configure VLAN gateways on Router 2

Connect Router 2's G0/0 to a trunk port on Switch 3.

```
Router2> enable
```

```
Router2# configure terminal
```

```
Router2(config)# interface gigabitEthernet 0/0
```

```
Router2(config-if)# no shutdown
```

```
Router2(config-if)# exit
```

```
Router2(config)# interface gigabitEthernet 0/0.30
```

```
Router2(config-subif)# encapsulation dot1Q 30
```

```
Router2(config-subif)# ip address 192.168.30.1 255.255.255.0
```

```
Router2(config-subif)# exit
```

```
Router2(config)# interface gigabitEthernet 0/0.40
```

```
Router2(config-subif)# encapsulation dot1Q 40
```

```
Router2(config-subif)# ip address 192.168.40.1 255.255.255.0
```

```
Router2(config-subif)# exit
```

3. Configure the router-to-router link

Use G0/1 between the two routers.

Router 1

```
Router1(config)# interface gigabitEthernet 0/1
```

```
Router1(config-if)# ip address 10.0.0.1 255.255.255.252
```

```
Router1(config-if)# no shutdown
```

```
Router1(config-if)# exit
```

Router 2

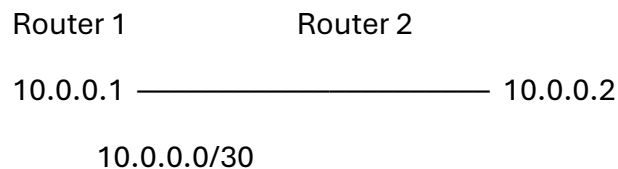
```
Router2(config)# interface gigabitEthernet 0/1
```

```
Router2(config-if)# ip address 10.0.0.2 255.255.255.252
```

```
Router2(config-if)# no shutdown
```

```
Router2(config-if)# exit
```

The link is therefore:



4. Add static routes

Router 1

Router 1 knows about VLAN 10 and VLAN 20 directly. Add routes to VLAN 30 and VLAN 40 through Router 2:

```
Router1(config)# ip route 192.168.30.0 255.255.255.0 10.0.0.2
```

```
Router1(config)# ip route 192.168.40.0 255.255.255.0 10.0.0.2
```

Router 2

Router 2 knows about VLAN 30 and VLAN 40 directly. Add routes to VLAN 10 and VLAN 20 through Router 1:

```
Router2(config)# ip route 192.168.10.0 255.255.255.0 10.0.0.1
```

```
Router2(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.1
```

5. Verify the routing table

On Router 1:

Router1# show ip route

You should see static routes marked with **S**:

S 192.168.30.0/24 [1/0] via 10.0.0.2

S 192.168.40.0/24 [1/0] via 10.0.0.2

On Router 2:

Router2# show ip route

Expected:

S 192.168.10.0/24 [1/0] via 10.0.0.1

S 192.168.20.0/24 [1/0] via 10.0.0.1

6. Configure PC default gateways

Example:

PC	VLAN	IP	Default Gateway
PC1	10	192.168.10.10	192.168.10.1
PC2	10	192.168.10.11	192.168.10.1
PC3	20	192.168.20.10	192.168.20.1
PC4	20	192.168.20.11	192.168.20.1
PC5	30	192.168.30.10	192.168.30.1
PC6	30	192.168.30.11	192.168.30.1
PC7	40	192.168.40.10	192.168.40.1
PC8	40	192.168.40.11	192.168.40.1

7. Test communication between VLANs

From **PC1 (VLAN 10)**:

ping 192.168.20.10

This tests VLAN 10 → VLAN 20.

Then:

ping 192.168.30.10

This tests VLAN 10 → VLAN 30 through the static route.

And:

ping 192.168.40.10

This tests VLAN 10 → VLAN 40.

A successful result should look like:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Screenshots for your practical record

Take these screenshots:

1. **Router 1:** show ip interface brief
2. **Router 2:** show ip interface brief
3. **Router 1:** show ip route
4. **Router 2:** show ip route
5. **PC1 → PC3:** successful ping between VLAN 10 and VLAN 20
6. **PC1 → PC5:** successful ping between VLAN 10 and VLAN 30
7. **PC1 → PC7:** successful ping between VLAN 10 and VLAN 40

Result: The static routes tell Router 1 how to reach VLANs 30/40 and Router 2 how to reach VLANs 10/20. Together with the VLAN gateways and trunk links, this enables communication between devices belonging to different VLANs.

5. Test end-to-end connectivity by pinging between devices in different VLANs and switches, and include screenshots of successful and failed ping results along with a brief explanation for any failures.

Hint: If pings fail, check VLAN assignments, trunk configuration, and static routes for errors.

-> End-to-End Connectivity Testing

After configuring the VLANs, trunks, and static routes, test communication between PCs that are on **different VLANs and different switches**.

Network used

PC	Switch	VLAN	IP Address
PC1	Switch 1	10	192.168.10.10
PC3	Switch 1	20	192.168.20.10
PC5	Switch 2	10	192.168.10.20
PC7	Switch 2	20	192.168.20.20
PC9	Switch 3	30	192.168.30.10
PC11	Switch 3	40	192.168.40.10

Test 1 — Different VLANs, same switch

From **PC1 (VLAN 10)**:

```
C:\> ping 192.168.20.10
```

Expected successful result:

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

 **Screenshot 1:** Capture the PC1 command prompt showing the successful ping.

Test 2 — Different VLANs and different switches

From **PC1 (VLAN 10, Switch 1)**:

```
C:\> ping 192.168.30.10
```

Expected:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

This verifies:

PC1

↓

VLAN 10

↓

Switch 1

↓

Trunk

↓

Router/static route

↓

Switch 2/3

↓

VLAN 30

↓

PC9

 **Screenshot 2:** Capture the successful PC1 → PC9 ping.

Test 3 — Another cross-VLAN test

From **PC5 (VLAN 10, Switch 2)**:

```
C:\> ping 192.168.40.10
```

Expected:

```
Reply from 192.168.40.10: bytes=32 time<1ms TTL=125
```

```
Reply from 192.168.40.10: bytes=32 time<1ms TTL=125
```

```
Reply from 192.168.40.10: bytes=32 time<1ms TTL=125
```

```
Reply from 192.168.40.10: bytes=32 time<1ms TTL=125
```

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)



Screenshot 3: Capture the successful PC5 → PC11 ping.

Failed Ping Test

For the practical demonstration, you can intentionally introduce an error and capture the failure.

For example, temporarily remove **VLAN 30** from the trunk:

```
Switch2(config)# interface gigabitEthernet 0/2
```

```
Switch2(config-if)# switchport trunk allowed vlan 10,20,40
```

Now PC1 attempts:

```
C:\> ping 192.168.30.10
```

Example failed output:

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.30.10:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

 **Screenshot 4:** Capture this failed ping.

Explanation of the failure

The ping fails because **VLAN 30 is not being carried across the trunk link**. The destination network cannot receive the traffic even though the PC's IP configuration may be correct.

Check:

show vlan brief

show interfaces trunk

show ip route

Restore VLAN 30:

Switch2(config)# interface gigabitEthernet 0/2

Switch2(config-if)# switchport trunk allowed vlan 10,20,30,40

Then repeat the ping. It should succeed if the remaining configuration is correct.

Troubleshooting checklist

Check	Command	What to verify
VLAN assignment	show vlan brief	PC port belongs to correct VLAN
Trunk	show interfaces trunk	Trunk is operational
Allowed VLANs	show interfaces trunk	Required VLAN is allowed
Interfaces	show ip interface brief	Interfaces are up/up
Static routes	show ip route	Remote VLAN networks have S routes
PC addressing	ipconfig	Correct IP, subnet mask, gateway
Connectivity	ping <IP>	End-to-end communication

Screenshot checklist for submission

1. **Successful:** PC1 → PC3, different VLANs.

2. **Successful:** PC1 → PC9, different VLANs and switches.
3. **Successful:** PC5 → PC11, different VLANs and switches.
4. **Failed:** PC1 → PC9 after intentionally removing VLAN 30 from the trunk.
5. **Verification:** show interfaces trunk.
6. **Verification:** show ip route.

Important: The ping outputs above are **sample expected results**, not screenshots from your actual equipment. For your lab report, run the commands on your physical PCs/switches and capture the real console/Command Prompt screenshots.